

Selection: if, if-else, nested if and switch ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write if, if-else, else-if ladder and nested if
- Write switch-case with break and default
- Draw the switch logical diagram
- Program: greater of two, pass/fail, leap year

```
if (marks ≥ 40)
    printf("Pass");
else
    printf("Fail");
```

```
if (n > 0) printf("Positive");
else if (n < 0) printf("Negative");
else printf("Zero");
```

switch (board long question)

```
int ch;
printf("1.Add 2.Sub 3.Mul\n");
scanf("%d", &ch);
switch (ch) {
    case 1: printf("Add"); break;
    case 2: printf("Sub"); break;
    case 3: printf("Mul"); break;
    default: printf("Invalid");
}
```

LEARN IT

switch works on int or char (not float). Each case should end with break, otherwise execution falls through. default is optional but expected in board answers.

Evaluate expression → Match case → Execute statements → break → leave switch

WORKED EXAMPLE / BOARD STYLE

Greater of two using a function (2026 Q2 xii):

```
int greater(int x, int y) {
    if (x > y) return x;
    else return y;
}
void main() {
    int a, b;
    scanf("%d%d", &a, &b);
    printf("Greater = %d", greater(a, b));
}
```

HOW BIEK ASKS THIS

2026 Section C Q3: Define switch with logical diagram. Draw a diamond or a multi-way box: expression, case 1, case 2, default.

اردو میں سمجھیں

if شرط سچی ہو تو بلاک چلتا ہے۔ switch ایک متغیر کی کئی قیمتوں پر چلتا ہے، ہر case کے بعد break لکھیں۔

CHECK YOURSELF

1. Can switch test a float?
2. What if you omit break?

Answers: 1. No · 2. Fall-through: the next case also runs

MCQs

1. switch is a:
 - a. loop
 - b. selection / decision structure
 - c. function
 - d. header
2. default in switch runs when:
 - a. always
 - b. no case matches
 - c. every case
 - d. at compile time

Key: 1=b, 2=b

Short questions (3 marks each)

1. Define switch with a small example.
2. Write a program that prints the greater of two integers using a function.
3. Differentiate if-else and switch.

Model short answers

1. switch(expression) { case c1: ... break; default: ... } selects one case by matching integers/chars.
2. Full greater() program as in the lecture.
3. if-else handles ranges and floats; switch is multi-way on discrete int/char values and is cleaner for menus.

Long questions (board Section C)

1. Define switch statement with a logical diagram and a complete C program (menu or grade).